



ABT RECERTIFICATION EXAMINATION

May 4, 2015

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 2006 and 2011. There are three parts to the exam. An answer sheet will be mailed for each part. These answer sheets must be returned to ABT, P.O. Box 97786, Raleigh, NC 27624 for grading by **July 24, 2015**. If you wish to return them by courier, the physical address is 1815 Oakboro Drive, Raleigh, NC 27614. If you do not achieve a passing score on any of the parts, you will be contacted and given another opportunity to pass. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the Recertification Committee's review of your other application materials. If you have any questions please contact the ABT office at (919) 841-5022 or info@abtox.org.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.
2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.
3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology, molecular toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet **ONLY** on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

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*Key for the following questions
Choose the single most appropriate answer*

1. Beryllium, a metal used in ceramic plants, the nuclear industry, and in space vehicles, can cause a severe adverse reaction in some people because of which of the following?

- A) it can cause a cell-mediated immune response
- B) it binds to metallothionein and accumulates in the kidney
- C) it reacts with sunlight to cause lesions on the skin
- D) these individuals lack detoxifying liver enzymes
- E) it can displace calcium in the bones, making them brittle and easy to break

2. Which of the following statements about factors impacting metal toxicity is NOT CORRECT?

- A) since metals can be quite reactive, the portal of entry is often initially the organ most affected
- B) for many inorganics there is no impediment to transplacental transport, therefore, human fetal blood metal levels are similar to maternal levels
- C) younger subjects are often less sensitive to metal intoxication due to less caloric consumption per pound of body weight than adults
- D) the essential metallic elements would be expected to show a "U"-shaped dose-response curve
- E) metal exposure can induce a cascade of molecular responses that may, in turn, reduce toxicity

3. Which of the following statements about mercury or methyl mercury is NOT CORRECT?

- A) methyl mercury crosses the placenta, and is concentrated to a level in fetal brain at least 5-7 times that of maternal blood
- B) consumption of predatory fish such as tuna and mackerel increases blood methyl mercury concentration
- C) the brain is the major target organ for inorganic mercury
- D) in severe cases of mercury poisoning with acute renal failure, hemodialysis may be the initial treatment along with chelation therapy
- E) inhalation of mercury vapor may produce an acute, corrosive bronchitis and interstitial pneumonitis

4. Toxicity from chronic exposure to cadmium is characterized by which of the following?

- A) calcium accumulation resulting in skeletal nodules
- B) cadmium-induced hypotension due to calcium channel blockade
- C) accumulation of cadmium-metallothionein complexes in liver mitochondria
- D) renal carcinoma originating in the glomeruli
- E) renal tubular dysfunction resulting in proteinuria

5. The toxicity of metals is often influenced by the valence state. Which of the following is true of the toxicology of chromium?

- A) chromium III has greater acute toxicity than Cr VI
- B) chromium III has greater chronic toxicity than Cr VI
- C) chromium VI is a lung carcinogen via inhalation while Cr III is not
- D) chromium III and chromium VI are approximately of equal toxicity
- E) chromium VI is less toxic when ingested than when inhaled

6. Which of the following BEST describes the metabolism of propylene glycol?

- A) glucuronidated and eliminated
- B) inactivated by epoxide hydrolase
- C) metabolized by alcohol dehydrogenase
- D) reduced by glutathione S-transferase
- E) not metabolized

7. Ethylene glycol is BEST described by which of the following?

- A) it is associated with metabolic alkalosis in humans following ingestion
- B) it is metabolized to oxalate and precipitated in the kidney as the Ca^{++} salt
- C) it is an excellent nontoxic solvent for many drugs
- D) it is primarily converted to a toxic metabolite by glutathione S-transferases
- E) it is metabolized to methylene glycol which can cause blindness

8. Ethanol can be given as an antidote for which of the following intoxications?

- A) phenobarbital
- B) phencyclidine
- C) iron
- D) ethylene glycol
- E) endrin

9. Exposure to which of the following compounds is MOST commonly associated with induction of leukemia in humans?
- A) asbestos
 - B) lead
 - C) benzene
 - D) paraquat
 - E) hexane
10. Which of the following are the most important aspects of pulmonary studies of single-walled carbon nanotubes?
- A) inclusion of biomarkers of liver injury to study designs
 - B) quality assurance auditing of 100% of the study data
 - C) collection of BAL fluid for analysis of cytokines and the presence of inflammatory cells
 - D) inclusion of a recovery phase three times the duration of the main study
 - E) physiochemical characterization of the test article and rigorous dose and endpoint controls
11. The insecticidal mode of action of the biopesticide *Bacillus thuringiensis* (Bt) is best described by which of the following?
- A) crystal Bt proteins form endotoxins that cause lysis of epithelial cells of the insect midgut
 - B) crystal Bt protein binds to and activates glutamate-dependent chloride channels of insect neurons
 - C) Bt proteins inhibit acetylcholinesterase, causing excess cholinergic stimulation
 - D) crystal Bt proteins bind to neuropathy target esterase and cause insect axonopathy
 - E) Bt proteins modify the kinetics of voltage-sensitive sodium channels in insect neurons
12. Pyrethroids are a class of insecticides that are widely used in agriculture and medicine. The effects of pyrethroids are mediated through which of the following mechanisms?
- A) inhibition of the enzyme cholinesterase
 - B) delayed inactivation of sodium channels in cells
 - C) delayed activation of potassium channels in cells
 - D) inhibition of ATPases
 - E) activating nicotinic acetylcholine receptors
13. Stimulation of which of the following receptors is responsible for the toxic action of carbamate insecticides?
- A) muscarinic
 - B) glutamate
 - C) glycine
 - D) serotonergic
 - E) dopamine
14. Which of the following chemicals bind acetylcholinesterase in a rapidly reversible manner?
- A) carbaryl
 - B) malathion
 - C) chlorpyrifos
 - D) diazinon
 - E) sarin
15. The mechanism of insecticidal activity of organophosphates is described by which of the following?
- A) anticoagulation
 - B) modification of voltage-sensitive sodium channels
 - C) inhibition of acetylcholinesterase
 - D) inhibition of the mitochondrial respiratory chain
 - E) activation of an octopamine-dependent adenylate cyclase
16. The platinum group of metals include ruthenium, rhodium, palladium, osmium, iridium, and platinum. Which of the following is TRUE regarding toxicity of metals in this group?
- A) osmium tetroxide is a powerful nephrotoxin
 - B) palladium chloride is readily absorbed from subcutaneous injection and is an important toxicant in industrial settings
 - C) platinum metal itself is a potent nephrotoxin
 - D) platinum salt sensitization is an acute effect rapidly fading after cessation of exposure
 - E) platinum compounds with antitumor activity produce proximal and distal tubular injury
17. A farmer came into the emergency department complaining of stomach pains, tingling in his extremities and hair loss. He indicated that he had been using a chemical found in his barn to kill rats. What chemical is the one most likely to have caused these symptoms?
- A) manganese
 - B) nicotine
 - C) rotenone
 - D) thallium
 - E) fluoroacetic acid
18. Acute organophosphate insecticide poisoning can be best treated with which of the following?
- A) methylene blue
 - B) vitamin K₁
 - C) atropine
 - D) ethanol
 - E) chelation therapy

19. The therapeutic efficacy of sodium nitrite in the treatment of patients with cyanide poisoning depends on which of the following?
- A) the formation of methemoglobin
 - B) an increase in oxidative phosphorylation
 - C) an increase in coronary arterial blood flow
 - D) a decrease in intracranial pressure
 - E) relaxation of the nonvascular smooth muscle
20. All of the following characteristics are factors in the toxicity of inhaled nanoparticles relative to larger particles EXCEPT which of the following?
- A) the biological nature of nanoparticles is associated with their chemical composition
 - B) nanoparticles may translocate from primary deposition sites to secondary organs
 - C) aggregation and agglomeration of nanoparticles are unlikely to affect the biokinetics
 - D) the biological nature of nanoparticles is associated with their physical form
 - E) nanoparticles within a biological environment can acquire a coating of protein which may affect the particle biodynamics
21. Carbon monoxide toxicity has been attributed to a reversible interaction with which of the following?
- A) myoglobin
 - B) cytochrome oxidase C
 - C) methemoglobin
 - D) hemoglobin
 - E) cytochrome P450
22. Insoluble particles that deposit in regions of the nose other than the anterior portion are mainly cleared by which of the following mechanisms?
- A) exhalation
 - B) absorption into the blood
 - C) carried by mucus to the glottis and then swallowed
 - D) phagocytosis by macrophages
 - E) cough
23. What range BEST defines the size constituting the definition of a nanoparticle?
- A) 1-5 μm
 - B) 0-100 nm
 - C) 0.01-0.1 μm
 - D) 1-5 nm
 - E) 10-100 nm
24. It is difficult to predict the toxicity of engineered nanomaterials based on their composition because of which of the following?
- A) they are inert substances in vitro and require specialized in vivo assays to assess
 - B) they have different physical and chemical properties than micro-scale products of the similar chemical composition
 - C) they are carcinogenic and dangerous to work with in the laboratory
 - D) handling them requires specific techniques beyond most testing laboratories capabilities
 - E) they are very unstable and their properties change drastically in a short period of time
25. After inhalation, which of the following is the most critical factor that will determine where along the respiratory tract that dust particles deposit?
- A) chemical composition of the dust
 - B) particle aerodynamic diameter
 - C) rate of particle dissolution
 - D) lining of the respiratory tract
 - E) particle clearance rate
26. Eating the leaves from foxglove plant can cause which of the following?
- A) cardiac arrhythmias
 - B) hepatotoxicity
 - C) renal toxicity
 - D) CNS stimulation
 - E) dermatitis
27. What is the toxic mechanism of action for venom from the Black Widow Spider?
- A) acts in the central nervous system by stimulating the pre-synaptic release of serotonin causing hallucinations
 - B) acts at the neuromuscular synaptic junction causing the release of neurotransmitters which causes muscle pain and cramps
 - C) acts by blocking optic nerve transmission causing temporary blindness
 - D) acts by stimulating histamine release from mast cells causing an anaphylactic reaction
 - E) acts by stimulation of the Purkinje fibers in the heart causing sudden cardiac death
28. What mechanism is chiefly responsible for the repair of non-bulky damage to DNA?
- A) direct repair
 - B) base excision repair
 - C) photolyase repair
 - D) recombinational repair
 - E) gene lesion repair

29. Which of the herbal products is NOT associated with toxicity of the organ identified?
- A) central nervous system: Kava Kava
 - B) hematological: Hops
 - C) hepatotoxins: Senna
 - D) respiratory: St. John's wort
 - E) cardiovascular system: European mistletoe
30. The venom in wasp stings (family *Vespidae*) works primarily through which of the following mechanisms?
- A) inhibition of acetylcholinesterase
 - B) release of histamine from mast cells
 - C) fibrinolytic enzyme activity
 - D) hemorrhagic metalloproteinase activity
 - E) induction of apoptosis
31. Which of the following is CORRECT regarding the Delaney clause?
- A) permits food additives to be declared GRAS
 - B) prohibits EPA from setting safe exposure levels to carcinogens
 - C) prohibits FDA from approving food additives found to cause cancer
 - D) only applies to food additives demonstrating human evidence of carcinogenicity
 - E) has resulted in a wide range of banned food additives
32. A mixture of melamine and cyanuric acid, neither of which is very toxic alone, caused the deaths of many pets and even human infants in 2007-08 when the mixture was found in animal feed ingredients and milk. What was the likely mode of action?
- A) acute cardiac toxicity
 - B) overwhelms homeostatic acid-base balance
 - C) forms co-precipitates in renal tubules
 - D) acute hepatic toxicity
 - E) blocks mitochondrial oxidative phosphorylation
33. Which of the following poses the greatest food-borne risk world-wide?
- A) chemical contaminants/adulterants (e.g., pesticides, hormones, antibiotics)
 - B) microbial contamination
 - C) contaminants and indirect additives from food processing/packaging
 - D) mycotoxins and food-borne molds
 - E) food additives
34. Acrylamide has been used in a variety of foods and beverages (e.g., baked goods and coffee, among others). Which of the following INCORRECTLY describes the properties of acrylamide?
- A) toxicity is primarily due to nervous system effects
 - B) acrylamide has been shown to be carcinogenic in animal studies
 - C) is slowly absorbed from the gastrointestinal tract but is rapidly excreted as a glucuronide conjugate
 - D) is formed at baking temperatures particularly in the presence of asparagine
 - E) the daily intake of acrylamide is about 200 times less than the NOAEL for nervous system effects
35. Which of the following toxic food contaminants are produced by cooking?
- A) botulinum toxin
 - B) tetrodotoxin
 - C) trichothecenes
 - D) heterocyclic amines
 - E) fumonisins
36. Clinical assessment of drug-induced liver injury using Hy's Law (indication of hepatocellular damage in conjunction with jaundice in the absence of cholestasis) considers which of the following test results?
- A) >3-fold increases in ALT, AST, and ALP without increased total bilirubin
 - B) >3-fold increases in ALT, AST, and ALP with >2-fold increases in total bilirubin
 - C) >3-fold increases in ALT and AST (not ALP) and >2-fold increases in total bilirubin
 - D) >3-fold increases in ALT, AST and >2-fold increases in total cholesterol
 - E) >25-fold increases in ALP and >2-fold increases in total bilirubin
37. There is scientific and regulatory attention being directed towards the increasing use of nanotechnology in many industries because of which of the following?
- A) nanoparticles are larger than microparticles
 - B) nanoparticles are most likely to impact the upper respiratory tract
 - C) nanoparticles are produced without being engineered to specific size or shape specifications
 - D) the surface chemistry and shape distributions of nanoparticles are unimportant in determining their effects
 - E) significant routes of exposure include inhalation, dermal and, possibly oral ingestion

38. Acetaminophen toxicity is best exemplified by which of the following?

- A) acetaminophen has been demonstrated to show a robust ($\geq 75\%$) inhibition of both COX-1 and COX-2
- B) hepatic necrosis is noted with acetaminophen overdose when saturation of glycine conjugation occurs resulting in the formation of a toxic metabolite
- C) the metabolic profile of acetaminophen *in vivo* involves bioactivation via the CYP2E1 pathway followed by multiple conjugation reactions
- D) N-acetylcysteine is contraindicated in treating individuals experiencing acetaminophen overdose (hepatic toxicity)
- E) renal tubular necrosis has not been associated with acetaminophen overdose in that the kidneys have the capacity to form mercapturic acid which readily eliminates the toxic parent compound

39. Excessive ingestion of which of the following food contaminants can elicit a wide range of clinical effects including acute gastrointestinal symptoms, effects on the central and peripheral nervous system and alopecia?

- A) vanadium
- B) aflatoxin
- C) tin
- D) thallium
- E) phthalates

40. Which of the following is NOT associated with paralytic shell fish poisoning caused by saxitoxin?

- A) suppression of AV nodal conduction
- B) Na^+ channel inhibition
- C) parathesia
- D) cyclopeptides
- E) asphyxiation

Key for the following questions
Choose the single most appropriate answer

1. Neoplastic transformation of cells is NOT usually associated which of the following?
 - A) activation of proto-oncogenes
 - B) increased growth factor production
 - C) inactivation of tumor suppressor genes
 - D) release of growth factors that suppress cell proliferation
 - E) inhibition of apoptosis
2. Which of the following molecules has NOT been classified as group 1 IARC carcinogenic agents?
 - A) arsenic
 - B) aflatoxin
 - C) caprolactam
 - D) benzene
 - E) vinyl chloride
3. Minocycline, nicardipine, and spironolactone are examples of marketed drugs known to induce thyroid neoplasia in rats by increasing circulating levels of TSH through which of the following mechanisms?
 - A) decreasing microsomal enzyme levels
 - B) increasing the release of T4 from thyroglobulin
 - C) microsomal induction resulting in decreased serum levels of T3, T4
 - D) chemical inhibition of 5'-monodeiodinase
 - E) increasing the organification of iodine through stimulation of thyroid peroxidase
4. What is the primary factor responsible for how deeply a gas penetrates into the lung?
 - A) vapor pressure
 - B) respiratory rate
 - C) water solubility
 - D) vapor density
 - E) molecular weight
5. In healthy individuals, mucociliary clearance of inhaled particles deposited in the lower airways of the tracheobronchial tree is generally completed generally within which of the following time periods?
 - A) 1-4 hours
 - B) 24-48 hours
 - C) 1 week
 - D) 3 weeks
 - E) 3 months
6. Which of the following assays assesses the ability of a test material to induce chromosomal aberrations *in vivo*?
 - A) bacterial-Ames assay (histidine reversion assay in *Salmonella*)
 - B) rodent micronucleus test
 - C) mammalian mouse lymphoma thymidine kinase assay
 - D) sister chromatid exchange assay
 - E) cytokinesis-block micronucleus assay
7. Which of the following is CORRECT regarding the "promotion" stage in the multistage carcinogenesis model?
 - A) it requires repeated application or continuous exposure to the test agent
 - B) it involves the conversion of benign preneoplastic lesions into neoplastic cancer
 - C) it is a rapid process that results in a carcinogen-induced mutational event
 - D) it is irreversible
 - E) it is not relevant to humans
8. During which period of pregnancy will administration of a drug most likely produce teratogenic effects on the skeletal system?
 - A) at the time of conception
 - B) first trimester
 - C) second trimester
 - D) third trimester
 - E) immediately before birth
9. How can DNA adducts assist in establishing a chemical's mode of action?
 - A) can interfere with accurate DNA replication, causing mutations
 - B) can interfere with the formation of the mitotic spindle, causing aneuploidy
 - C) can cause protein adducts that lead to cytotoxicity
 - D) predict that a chemical will be carcinogenic *in vivo* but will have no effects *in vitro*
 - E) act as a secondary messenger, causing a change in transcription rates
10. Which of the following is CORRECT about fibrosarcomas?
 - A) they typically occur in tissues of mesenchymal origin
 - B) they are characteristically benign
 - C) they are the most common liver tumors in Sprague-Dawley rats
 - D) they differ from benign fibromas only in size
 - E) prepared slides of fibrosarcomas are virtually devoid of mitotic figures

11. Which of the following statements is NOT TRUE regarding chemical-induced deregulation to produce thyroid hormone insufficiency?

- A) there are multiple sites within the pituitary-thyroid axis where chemical interaction may cause de-regulation
- B) liver UDP-glucuronyltransferase induction may lead to thyroid hormone insufficiency in rats
- C) humans are more sensitive to thyroid hormone de-regulation than laboratory rat models
- D) thyroid follicular cell cancer may result from chronic thyroid hormone insufficiency in laboratory rats
- E) thyroid insufficiency can impact normal brain development

12. What is the most critical factor in the site of deposition within the respiratory tract for an inhaled particle?

- A) respiratory rate
- B) solubility of the particle
- C) particle size
- D) shape of the particle
- E) tidal volume

13. Which of the following is considered a reproductive physiology similarity among humans and rats?

- A) the menstrual cycle has a spontaneous luteal phase of 10 to 14 days
- B) FSH and LH regulated germ cell development after puberty
- C) pregnancy is easily disrupted by estrogens
- D) placental tissue expresses high levels of aromatase
- E) the spermatogenic cycle is approximately 75 days in duration

14. Which of the following is NOT CORRECT regarding endocrine disruption?

- A) at birth, anogenital distance and nipple retention should be evaluated when testing for endocrine disruption, because these effects are highly correlated with malformations
- B) antiandrogens cause masculinization of female fish
- C) endocrine disruption noted in fish near a human sewage effluent source may be the result of exposure to ethanyl estradiol
- D) the mode of action for male reproductive toxicity of dibutylphthalate involves interaction with the androgen receptor
- E) endocrine disrupting chemicals act through mechanisms that include binding to the androgen receptor, estrogen receptor, retinoic acid receptor, inhibition of steroidogenic enzymes or the synthesis of thyroid hormones

15. Ethanol, retinoids, valproic acid, and the angiotensin converting enzyme (ACE) inhibitors all have which of the following effects?

- A) cause liver toxicity
- B) lower blood pressure
- C) are human developmental toxicants
- D) have central nervous system effects
- E) have similar pharmacokinetic pathways

16. Which of the following is the biggest public health concern related to low level exposure to lead (e.g. blood level)?

- A) encephalopathy
- B) proximal tubular nephropathy in the elderly
- C) anemia
- D) cognitive delays and behavioral changes in children
- E) hemoglobin synthesis

17. The basis of acrylamide neurotoxicity is considered to be which of the following?

- A) segmental demyelination
- B) necrosis of basal ganglia
- C) axonal degeneration
- D) neurofibrillar aggregates
- E) enlarged astrocytes

18. Which of the following is TRUE regarding glutamate?

- A) it is the excitatory neurotransmitter in the brain
- B) its effects are mediated by ionotropic (coupled with G proteins) and metabotropic (ligand-gated directly to ion channels) receptors
- C) its entry into the CNS is not regulated at the blood brain barrier
- D) it is neurotoxic at normal concentrations in the brain
- E) it is toxic to dendrites, neuronal cell bodies and axons

19. Which of the following is NOT CORRECT regarding peripheral nervous system axonopathies?

- A) the axon degenerates and with it the myelin surrounding the axon; however, the cell body of the neuron remains intact
- B) the cell body of the neuron is the focus of toxicity and the distal axon degenerates from the synapse back toward the cell body
- C) following a chemical induced transection of the axon at some point along its length, the axon distal to the transection degenerates
- D) longer axons have more targets and are more vulnerable than shorter axons to damage by toxic agents
- E) the cell body of the neuron undergoes a process of chromatolysis in response to a chemical induced transaction of the axon

20. The potential effects of a toxicant on vision is evaluated in the Functional Observational Battery in which of the following ways?
- A) evaluation of ocular irritancy
 - B) observing the response of the pupil to light
 - C) evaluation of rod and cone density
 - D) scoring the degree of lens opacity
 - E) evaluation of pattern-elicited visual evoked potentials
21. The peripheral neuropathy caused by n-hexane and carbon disulfide is best associated with which of the following?
- A) axonal degeneration
 - B) cholinesterase inhibition
 - C) formation of Heinz bodies
 - D) immunosuppression
 - E) decreased dopamine level
22. The mechanism of retinal and optic nerve toxicity of methanol in humans is BEST attributed to which of the following?
- A) biotransformation to formaldehyde and subsequent reaction with optic nerve proteins
 - B) photo-oxidation in the vitreous humor and generation of reactive oxygen species
 - C) biotransformation to formic acid and subsequent disruption of oxidative energy metabolism
 - D) the presence of catalase in the optic nerve, leading to high levels of intracellular hydrogen peroxide
 - E) condensation with intraocular melanin to form a photosensitive by-product
23. Which of the following is NOT a characteristic of Natural Killer (NK) cells?
- A) derived from bone marrow
 - B) produces interferon gamma
 - C) major function is cytolysis
 - D) possess immunologic memory
 - E) is a cellular component of the innate immune system
24. Which of the following is NOT CORRECT regarding renal injury?
- A) detachment of podocytes from the glomerular basement membrane is thought to be a basis for the decrease in size selectivity for compounds excreted in urine
 - B) alterations of concentrating ability are one characteristic of compounds that damage more distal structures in the nephron
 - C) the proximal tubule has a key role in transport and reabsorption processes; regional differences in transport function are not solely responsible for regional differences in susceptibility
 - D) exposure may mediate some of the susceptibility of this tissue because kidneys constitute approximately one-half percent of body mass and receive approximately 20% of the cardiac output
 - E) the glomerulus is the most common site of renal injury
25. Which of the following is the primary site of kidney injury resulting from exposure to chloroform?
- A) proximal tubule
 - B) glomerulus
 - C) loop of Henle
 - D) distal tubule/collecting duct
 - E) medulla/papilla
26. Which of the following statements BEST characterizes the Buehler and Maximization assays?
- A) they are conducted to evaluate the potential of a product to produce skin irritation
 - B) they are most commonly performed in minipigs
 - C) they require 72 hour to perform
 - D) they evaluate delayed-contact hypersensitivity
 - E) they are *in vitro* assays for cell-mediated immune response
27. Chloracne is a persistent skin disease caused by overexposure to certain halogenated aromatic hydrocarbons like 2,3,7,8-tetrachlorodibenzo-p-dioxin. What is the prominent target tissue for chloracnegenic agents within the skin matrix?
- A) stratum corneum
 - B) sebaceous (oil) glands
 - C) viable hypodermis
 - D) sweat glands
 - E) subcutaneous adipose

28. Which of the following is NOT characteristic of allergic contact dermatitis?
- A) low molecular weight chemicals (haptens) are most often responsible
 - B) minute quantities of an allergenic chemical may elicit an overt reaction
 - C) sensitization must occur first
 - D) the intensity of the reaction is proportional to the dose applied
 - E) contact sensitivity to an agent that is systemically administered may cause symptoms such as headache, malaise, and arthralgia
29. Which ocular tissue is highly vulnerable to systemic, toxicant induced structural and/or functional damage?
- A) iris
 - B) lens
 - C) cornea
 - D) ciliary body
 - E) retina
30. The oral anticoagulant, warfarin, modulates hemostasis by interfering with vitamin K metabolism resulting in
- A) increased fibrinolytic activity
 - B) decreased vitamin K epoxide levels
 - C) decreased fibrogen levels
 - D) decreased clotting factors IX and X
 - E) thrombocytopenia
31. Which of the following agents specifically accumulates in hepatic stellate cells?
- A) iron
 - B) vitamin A
 - C) endotoxin
 - D) acetaminophen
 - E) pyrrolizidine alkaloids
32. Which of the following is the primary function for Kupffer cells in the liver?
- A) support the cords of hepatocytes in the liver
 - B) macrophages that ingest and degrade particulate matter
 - C) provide support to bile ducts
 - D) play a major role in the formation of bile
 - E) provide nutrient homeostasis
33. The *in vivo* micronucleus test is often used in a standard battery of genetic toxicity assays. In this test, the micronuclei are defined as which of the following?
- A) pyknotic nuclei from a cell with decreased cytoplasmic to nuclear ratios
 - B) clumps of ribosomes
 - C) membrane-bounded structures that contain chromosomal fragments or whole chromosomes
 - D) binucleated remnants of germ cells
 - E) dense bodies of lysosomal origin
34. Which of the following answers is CORRECT regarding IARC Class 1 carcinogens with a sufficient evidence (causal relationships) for human carcinogenicity?
- A) arsenic, benz(a)anthracene, diethylnitrosamine, urethane, estrogens
 - B) aflatoxin, benzene, 5-Azacytidine, estrogens
 - C) benzene, aflatoxin, vinyl chloride
 - D) TCDD, arsenic, urethane
 - E) alcohol, estrogen, testosterone
35. The tester strains of *Salmonella typhimurium* used in the Ames Assay are which of the following?
- A) TA 1535, TA 100, TA 1538, TA 98
 - B) TA 1540, TA 200, TA 1520, TA 80
 - C) TA 1548, TA 202, TA 1505, TA 92
 - D) TA 1536, TA 110, TA 1518, TA 85
 - E) TA 1510, TA 105, TA 536, TA 98
36. Which organ system is most susceptible to cisplatin toxicity?
- A) kidney
 - B) liver
 - C) CNS
 - D) heart
 - E) eye
37. Which of the following associations is NOT CORRECT?
- A) mercury : acute renal failure in humans
 - B) mercury : acute renal failure in rats
 - C) unleaded gasoline : α_{2u} -globulin in humans
 - D) unleaded gasoline : α_{2u} -globulin in rats
 - E) cadmium : B₂-microglobulin in humans

38. Iron (Fe) is an essential mineral but overdoses result in damage to periportal hepatocytes. Which of the following best describes the reason for this specificity?

- A) excess Fe from the portal vein precipitates in the zone I hepatocytes causing mechanical disruption of the cells, and the concentration of Fe in the zone I
- B) Fe functions as an electron donor for the formation of reactive oxygen species leading to lipid peroxidation and the concentration of oxygen available to participate in this reaction is higher in the zone I region of the liver compared to the zone III region
- C) Fe is preferentially taken up by the zone I hepatocytes which overwhelms the ability of the cells to regenerate ATP via normal respiratory mechanisms
- D) excess Fe directly interferes in the formation of the cytoskeletal structure leading to inhibition of spindle formation and eventual loss of mitotic capability
- E) Fe is preferentially taken up in type I hepatocytes and inhibits ATP-dependent formation of bile leading to cholestasis in the zone I region

39. Xenobiotics such as omeprazole or zidovudine can contribute to a deficiency of vitamin B₁₂. Subsequently, this deficiency can result in changes in erythroid and myeloid lineages in bone marrow. These events can result in what type of anemia or condition?

- A) aplastic anemia
- B) sideroblastic anemia
- C) megaloblastic anemia
- D) iron-deficiency anemia
- E) pure red cell aplasia

40. Of the various circulating biomarkers available for detecting myocardial injury, which one of the following is considered to be predominately expressed in cardiomyocytes?

- A) B-type natriuretic factor (BNP)
- B) cardiac troponins T and I (cTnT and cTnI)
- C) C-reactive protein (CRP)
- D) creatine kinase-MM (CK-MM)
- E) myoglobin

Key for the following questions
Choose the single most appropriate answer

1. Clinical chemistry profiles that show large increases in alanine aminotransferase (ALT), sorbitol dehydrogenase (SDH) and aspartate aminotransferase (AST) activities commonly relate to which of the following?
 - A) are relatively non-specific indicators of toxicity
 - B) often would be accompanied by necrosis in the kidney
 - C) would not be biologically plausible
 - D) would be an expected response to a liver toxicant
 - E) represent leakage of enzymes from damaged blood cells into the plasma
2. The Therapeutic Index (TI) of a drug is an approximation of the relative safety of a drug and is expressed as the ratio of the adverse or toxic dose (TD50) to which of the following?
 - A) lethal dose (LD)
 - B) no observable adverse effect level (NOAEL)
 - C) margin of safety (MOS)
 - D) median effective dose (ED50)
 - E) margin of exposure (MOE)
3. Which of the following best describes hormesis?
 - A) beneficial responses at low doses and adverse effects at high doses
 - B) teratogenicity due to estrogenic activity
 - C) metabolic alteration resulting in hormone change
 - D) increase in hormone production
 - E) pathological change in sex organs
4. Primary sites of absorption of toxicants into the body include which of the following?
 - A) skin, gastrointestinal tract, eyes
 - B) skin, gastrointestinal tract, hair
 - C) skin, gastrointestinal tract, lung
 - D) lung, gastrointestinal tract, eyes
 - E) skin, lung, eyes
5. What reactive oxygen species (ROS) is likely to be most damaging to an organism?
 - A) hydrogen peroxide (H_2O_2)
 - B) metal oxides
 - C) hydroxyl radical ($OH^\cdot$)
 - D) hydroxyl ion (OH^-)
 - E) nitric oxide ($NO^\cdot$)
6. The cytochrome P450 family of enzymes consists of about 1,000 known P450 isoforms; about 50 of these are functional in humans. What factors may play a role in the susceptibility of individuals to the metabolic activation of procarcinogenic environmental carcinogens, such as some polycyclic aromatic hydrocarbons found in tobacco smoke, to the ultimate carcinogenic form?
 - A) the presence of CYP3A4 and CYP3A5 isoforms in the intestine and kidney
 - B) the inhibition of CYP3A by a constituent of grapefruit juice
 - C) polymorphism of the NAT2 (N-acetyltransferase) conjugating enzyme
 - D) the presence of isoforms such as CYP1A1/2, CYP2A6, CYP2B1, and CYP2E1
 - E) the presence of conjugating enzymes such as glucuronosyltransferases
7. Which of the following enzymes catalyze a phase II biotransformation of xenobiotic substances?
 - A) epoxide hydrolase
 - B) glutathione S-transferase
 - C) monoamine oxidase
 - D) UDP-glucose pyrophosphorylase
 - E) xanthine oxidase
8. Peanut allergy is relatively common and can be quite severe in some individuals. Although food allergies that develop in childhood may be outgrown, peanut allergy is often lifelong. The primary mechanism of hypersensitivity to peanuts is which of the following?
 - A) Type 1 hypersensitivity
 - B) Type 2 hypersensitivity
 - C) Type 3 hypersensitivity
 - D) Type 4 hypersensitivity
 - E) Type 5 hypersensitivity
9. Which of the following metabolic pathways would most likely be involved with protecting tissue and cellular function damage by highly electrophilic metabolites of xenobiotics?
 - A) acetylation
 - B) sulfation
 - C) glutathione conjugation
 - D) methylation
 - E) amino acid conjugation

10. Which of the following statements is NOT CORRECT relative to toxicokinetic behavior of pharmaceuticals?

- A) xenobiotics eliminated in vivo via first-order processes are essentially completely eliminated ($\geq 99\%$) over seven (7) half-lives
- B) a high apparent volume of distribution is indicative that the xenobiotic predominantly remains in plasma
- C) clearance of a xenobiotic from the body may encompass renal or biliary excretion, pulmonary exhalation, and/or hepatic biotransformation
- D) bioavailability describes the extent of absorption via extravascular administration to that determined following intravenous administration by a comparison of AUC values
- E) in a two-compartment model of a xenobiotic, a semi-logarithmic plot of plasma concentration vs. time demonstrates a biphasic slope

11. Which of the following event is not implicated in the process of apoptotic cell death?

- A) cell shrinking
- B) condensation of nuclear material
- C) formation of membrane-bound cellular fragments
- D) inflammatory reaction
- E) phagocytic activity

12. A common hematological manifestation of lead intoxication is a hypochromic microcytic anemia especially in children. The anemia is thought to result from a decreased life span of the erythrocytes and an inhibition of heme synthesis. What are the two major enzymes that are known to be inhibited by lead?

- A) δ -aminolevulinate synthase, δ -aminolevulinic acid dehydratase
- B) δ -aminolevulinate dehydratase, ferrochelatase
- C) δ -aminolevulinic acid dehydratase, uroporphyrinogen decarboxylase
- D) uroporphyrinogen decarboxylase, δ -aminolevulinic acid dehydratase
- E) uroporphyrinogen decarboxylase, ferrochelatase

13. The principle application of PBPK models is for which of the following?

- A) determine the appropriate species for testing a pesticide ingredient
- B) assess structure activity relationship between a group of chemicals
- C) predict the target tissue dose of the toxic parent chemical or its reactive metabolites
- D) interpret gene array data
- E) serve as a basis for the application of intraspecific uncertainty factors for cancer vs. non-cancer risk assessment

14. For drugs that are extensively bound to plasma proteins but are not bound to tissue components, which of the following is most correct ?

- A) renal clearance will be equivalent to hepatic clearance
- B) clearance will be less than the half life
- C) volume of distribution will approach that of plasma volume
- D) volume of distribution will be dependent upon hematocrit
- E) hepatic clearance will be greater than renal clearance

15. The transcription factor Nrf2 is the master regulator of a major adaptive response to cellular stress resulting in induction of detoxification enzymes. The protein which anchors Nrf2 in the cytosol and dissociates from Nrf2 under stress conditions is which of the following?

- A) glutathione reductase
- B) glutathione peroxidase
- C) p53
- D) Keap 1
- E) TNF

16. Within the risk assessment framework, evaluation of the relevant mode of action is key to risk characterization. Certain carcinogens have mechanisms that result in tumors with no relevance to humans. An example of this is which of the following?

- A) benzene exposure and malignant lymphomas in mice
- B) nickel oxide exposure and lung tumors in hamsters
- C) unleaded gasoline exposure and renal tumors in male rats
- D) carbon tetrachloride exposure and liver tumors in rats
- E) hexavalent chromium exposure and lung carcinomas in mice

17. There is a growing interest in nanoparticles that may be useful in a variety of chemical and biological applications. Which statement below BEST describes our current knowledge of the technology?

- A) nanoparticles are usually described as particles 300nm (diameter) or smaller in size
- B) nanoparticles follow the same principles of absorption, distribution, metabolism and excretion as larger particles
- C) one of the toxicological concerns of nanoparticles is the enormous surface area relative to their mass
- D) FDA regulates nanotechnology no differently from other technology
- E) it had been demonstrated that intratracheally administered nanoparticles behave similarly as larger particles

18. John really enjoys grapefruit juice and consumes several large glasses per day. He recently started taking a new medication and has begun to experience severe muscle aches. Lab tests have revealed elevated creatine kinase and alanine aminotransferase levels in John's blood. Which medication is he likely taking?

- A) cyclosporine
- B) atorvastatin
- C) naproxen
- D) acetaminophen
- E) hydrochlorothiazide

19. Cerivastatin is a cholesterol-lowering drug recalled from the market due to many adverse reactions and over 50 deaths mainly due to rhabdomyolysis, a potential adverse response with any statin drug. What is rhabdomyolysis?

- A) liver failure due to cholesterol deficiency
- B) toxicity to the small intestine
- C) primary muscle damage which causes secondary kidney failure
- D) a direct toxic effect on the lungs which causes secondary muscle damage
- E) pancreas failure which causes brain damage due to hypoglycemia

20. Which of the following is an anti-apoptotic protein?

- A) Bax
- B) Bak
- C) Bad
- D) Bcl-2
- E) Bid

21. The key measures of validity in epidemiologic studies are sensitivity and specificity. Which of the following statements is NOT correct?

- A) sensitivity is a measure of how well the test identifies a true condition (disease or exposure)
- B) specificity is a measure of how well it documents a true non-condition (the absence of disease or exposure)
- C) consistency (i.e., precision) of measurement does not ensure the absence of measurement bias
- D) precision of measurement is more important than accuracy
- E) the operational diagnostic breakpoint between normal and abnormal individuals can impact both the sensitivity and specificity of a test

22. The primary physiochemical property used in quantitative structure-activity relationship (QSAR) models to predict bioconcentration in fish is which of the following?

- A) Henry's Law constant squared
- B) log octanol/water partition coefficient
- C) affinity to organic carbon (Koc)
- D) water solubility
- E) molecular weight

23. If the total daily intake of a specific chemical agent via drinking water is 0.1 mg/kg/day, and the NOAEL (no observed adverse effect level) for neurotoxicity is 100 mg/kg/day, which of the following is the MOE (margin of exposure) for neurotoxicity via the oral route?

- A) 10
- B) 0.001
- C) 10 mg/kg/day
- D) 1000
- E) MOE cannot be calculated without an uncertainty factor (UF)

24. What are the four key steps in the risk assessment process?

- A) research, hazard assessment, exposure calculation, risk characterization
- B) research, dose-response assessment, exposure assessment, risk communication
- C) hazard identification, exposure assessment, risk characterization, risk management
- D) hazard identification, dose-response assessment, exposure assessment, risk characterization
- E) problem formulation, dose-response assessment, risk characterization, risk communication

25. Drug induced QT interval prolongation and proarrhythmia (induction of torsade de pointes (TdP)) is a mechanism based cardiac toxicity that is most commonly due to which of the following?
- A) inhibition of IKr, a potassium current
 - B) inhibition of ICa, a calcium current
 - C) inhibition of INa, a sodium current
 - D) inhibition of ICl, a chloride current
 - E) activation of IKr, a potassium current
26. Epidemiological data may be considered to be preferable for dose-response modeling, primarily because of which of the following?
- A) the causal relationship between dose and response can be easily established
 - B) consideration of the impact of other variables on the outcome is generally not necessary
 - C) the impact of exposure over a lifetime can be more easily examined than in a multiyear laboratory animal study
 - D) it is possible to model dose-response relationships directly in humans, thereby avoiding concern for species differences
 - E) by using biomarkers, the doses associated with effects can be more precisely defined, thereby avoiding the need for external dose estimation
27. In an aquatic environment, which information is LEAST important to fully evaluate a marketed drug's environmental risk?
- A) mode of action of the drug
 - B) partition coefficient of the drug in octanol/water
 - C) the drug's NOEC (No Observed Effect Concentration) from long-term toxicity tests with fish
 - D) the Predicted Environmental Concentration (PEC) of the marketed drug in surface water
 - E) the drug's ability to undergo biodegradation
28. Biomarkers in ecotoxicology provide information on which of the following criteria?
- A) ecosystem stability and resilience
 - B) species richness and evenness
 - C) organismal exposure and effect
 - D) community structure and function
 - E) trophic relationships and energetics
29. Phthalates are often classified as endocrine disruptors or hormonally-active agents (HAAs) because of their potential ability to interfere with the endocrine system. Significant exposure to phthalates in children may result from which of the following?
- A) consumption of fruits and vegetables treated with pesticides
 - B) exposure to air emissions from coal-fired power plants
 - C) incidental ingestion of urban soils while playing outdoors
 - D) mouthing of plastic toys
 - E) inhalation of exhaust from diesel engines
30. During aquatic toxicology testing, the test organism utilized as a sentinel species for aquatic invertebrates is which of the following?
- A) *Skeletonema costatum*
 - B) *Microcystis aeruginosa*
 - C) *Daphnia magna*
 - D) *Pimephales promelas*
 - E) *Oncorhynchus mykiss*
31. The effluent from a municipal sewage plant induces plasma vitellogenin in trout. Which of the following is MOST likely the cause?
- A) conjugated estrogens
 - B) cadmium
 - C) 2,3,7,8-tetrachlorodibenzo-p-dioxin
 - D) low oxygen tension
 - E) malathion
32. Which of the following BEST describes why inhalation of long asbestos fibers (>10um) is hazardous in the lung?
- A) long fibers have a smaller surface area
 - B) long fibers are known to produce asbestosis
 - C) incomplete phagocytosis of long fibers by macrophages prevents lung clearance
 - D) long fibers penetrate to the lung periphery where they become lodged
 - E) long fibers are readily transported to the pleura via lymphatics
33. Studies of asthma prevalence in occupational settings with repeated exposure to low-molecular weight agents have suggested exposure to which of the following?
- A) vinyl chloride
 - B) toluene diisocyanate
 - C) 2-naphthylamine
 - D) benzidine
 - E) copper

34. Elimination half-life ($T_{1/2}$) of a toxicant increases in proportion to which of the following?
- A) total body clearance
 - B) dose
 - C) absorption
 - D) volume of distribution
 - E) hepatic blood flow
35. Which route of exposure is the primary concern in the vast majority of work environments?
- A) oral
 - B) respiratory
 - C) dermal
 - D) ocular
 - E) injection
36. Two-dimensional polyacrylamide gel electrophoresis provides elucidation of what molecular components?
- A) cDNA for transcriptional profiling
 - B) nucleotide mutations
 - C) DNA adducts
 - D) small molecule chemical metabolites in the urine
 - E) proteins expressed in fluids or tissues
37. Which of the following steps is NOT deemed critical to the process of risk assessment?
- A) hazard identification
 - B) dose-response assessment
 - C) exposure assessment
 - D) risk characterization
 - E) risk management
38. The initiation stage in the carcinogenesis process is characterized by which of the following?
- A) selection of clonal expansion of initiated cells to produce a preneoplastic lesion
 - B) rapid, irreversible process resulting in carcinogen-induced mutational event
 - C) conversion of benign neoplastic lesions into neoplastic cancer
 - D) nongenotoxic event resulting in sustained cell proliferation and/or inhibition of apoptosis
 - E) single event that is sufficient for neoplastic formation
39. Tumor suppressor genes and oncogenes play a role in cancer induction. Which of the following is NOT CORRECT about tumor suppressor genes and oncogenes?
- A) the tumor suppressor p53 is responsible for arresting cell replication
 - B) the tumor suppressor p53 functions to promote apoptosis which can prevent proliferation of mutated cells
 - C) mutation of the BRCA1 tumor suppressor gene is linked to the development of breast cancer
 - D) oncogenes and tumor suppressor genes are important for transformed cells *in vitro* but do not influence cancer development *in vivo*
 - E) oncogenes and tumor suppressor genes encode proteins that control cell growth and proliferation
40. Which of the following BEST matches a particular type of lesion with the genotoxic agent most likely to create it?
- A) alkylation - ionizing radiation
 - B) pyrimidine dimers - ultraviolet radiation
 - C) double-strand breaks - exposure to chemicals such as 9-aminoacridine
 - D) intrastrand cross links - endogenous oxygen radicals
 - E) intercalations - exposure to chemicals such as benzo(a)pyrene